

Proton Beam Compression Studies in the Spallation Neutron Source for a 10 TeV Muon Collider

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This paper represents my work in accordance with University regulations.

/s/ Inci Karaaslan

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1 Introduction

More than a decade after the discovery of the Higgs Boson at the Large Hadron Collider at CERN, the new physics implications of the Higgs Boson, among other Beyond the Standard Model (BSM) observations, motivate the construction of new particle colliders to extend the reach of current proton-proton accelerators. Future collider R&D is motivated by two main goals: the first is greater-precision studies of the dynamics of already discovered particles such as the Higgs boson in the Standard Model (SM), while the second involves going to energy-mass scales in the tens of TeV to directly search for new particles and phenomena predicted by various BSM models such as SUSY [1]. To achieve the first goal, efficient e^-e^+ colliders otherwise known as particle factories or Higgs factories when they operate with a total energy of 240-250 GeV available for particle production in the center-of-mass frame (referred to directly as center-of-mass energy or c.m.e. hereafter) are designed, producing high rates of Higgs bosons to measure of Higgs couplings to fermions and gauge bosons as well as self-coupling and rare decays. On the other hand, there are various collider designs with a selection of candidate colliding particles to reach higher and higher energy scales to discover new particles and achieve the second goal. One of the most promising candidates for these energy-frontier colliders is the 10 TeV c.m.e. $\mu^-\mu^+$ Collider.

The Spallation Neutron Source (SNS) at Oak Ridge National Laboratory (ORNL) is uniquely positioned to develop the required R&D and tackle potential challenges that arise in the Muon Collider since the bunch formation steps in the front-end proton driver are closely analogous to the operation at SNS. One of the critical steps is compression of a relatively long initial proton bunch to its rather short final length, which is the focus of this paper. More specifically, the project discussed in this paper aims to benchmark and improve the current proton bunch compression experiments at the SNS for a 10 TeV Muon Collider and explore the unique space-charge regime due to the high-intensity proton beam at the SNS. To this end, experimental studies at the SNS Control Room were conducted and compared to pyORBIT simulations with varying physical parameters and effects.

This section aims to provide the motivation behind a muon collider, some of the accelerator physics challenges, as well as the basic physical intuition required to understand the parameters involved in experiments and simulations at the SNS, discussed in subsequent sections. The next section will provide a general overview of the Spallation Neutron Source as it relates to the proposed 10 TeV Muon Collider and outline the project goals and conduct. Finally, the last section will discuss the results obtained from experiment and the subsequent

simulation.

1.1 A 10 TeV Future Muon Collider

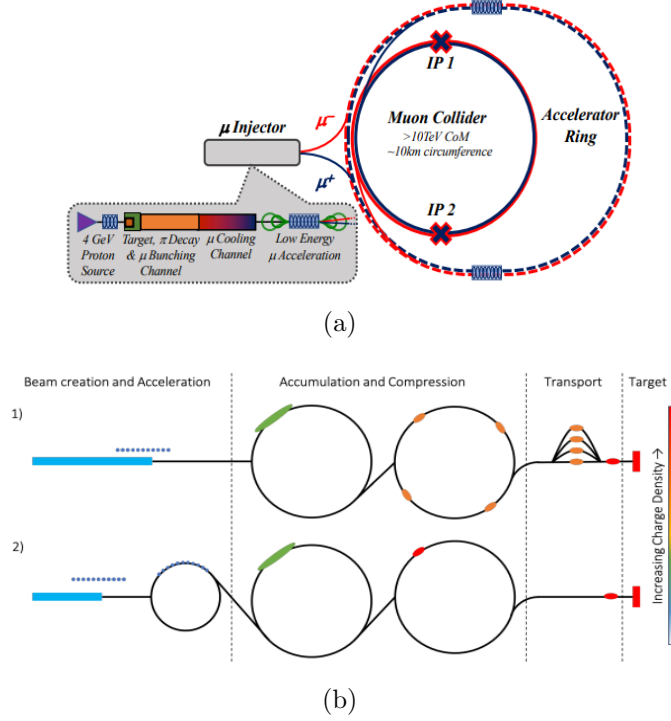


Figure 1: (a) Muon collider conceptual overview [2]. The μ Injector component includes the proton driver, the complex from a 4 GeV p^+ source to the target that facilitates the production of pions, pion decay channel, and muon cooling within the μ Injector Complex. (b) proton driver from (a) in greater detail. The proposal for the Proton Complex [3] includes two different schematics, where 1 includes an accumulation ring and a separate compression ring and short single bunches extracted from the compressor getting recombined, while 2 includes an additional ring and works not in short single bunches but a train of bunches that were combined before the compressor. The main proposal concerns the first schematic.

Muons, and subsequently muon colliders, combine some of the key advantages that come with e^-e^+ and p^+p^+ colliders. Namely, muon colliders can be circular and much smaller than current designs of linear electron colliders and hadron colliders with the same effective energy, making them one of the primary candidates for extension to higher energies. When compared with electron colliders, this can be attributed to energy loss due to synchrotron radiation, radiation emitted by a charged particle when its acceleration is perpendicular to its velocity, just like when colliding particles go around a circular collider. The power carried

by synchrotron radiation is found by the relativistic Larmor formula,

$$P = \frac{q^2}{6\pi\epsilon_0 c^3} a^2 \gamma^2 = \frac{q^2 c}{6\pi\epsilon_0} \frac{(\beta\gamma)^4}{\rho^2} \propto \left(\frac{E}{mc^2}\right)^4 \quad (1)$$

where ϵ_0 is the vacuum permittivity, q is the particle charge, a is the magnitude of the acceleration, c is the speed of light, γ is the Lorentz factor, $\beta = v/c$, ρ is the radius of curvature of the particle trajectory, and E is the relativistic energy of the particle while m is the mass of the particle [4]. Given that $m_\mu/m_e \approx 200$, the power loss of an electron due to synchrotron radiation is over a billion times bigger. When compared with hadron colliders, this can be attributed to the fact that, unlike electrons and muons, protons are not fundamental particles but quark-gluon composite objects, which means that not all of the beam energy is available to the individual quarks that make up the proton and interact with one another during collision. For example, a 10 TeV muon collider could allow exploration of similar physics as a 100 TeV proton collider [5].

As a potential future collider with a goal to probe energy scales higher than the current Large Hadron Collider (LHC) in CERN, the Muon Collider shares similar core considerations when it comes to collision energies and intensities in interaction points (IPs) as well as overall integrated luminosities. While reaching increasing energy scales is essential for energy-frontier colliders generally, a sufficiently large number of detectable interactions is also needed for full exploration of BSM sectors. The number of events of interest N can be defined as:

$$N = \sigma \int \mathcal{L}(t) dt \quad (2)$$

where σ is the cross-section of the relevant particle interaction and $\mathcal{L}$ is the instantaneous luminosity, the number of events detected in a short period dt within a cross-section. For colliders, the luminosity, and thus the number of events of interest, is defined only by the beam parameters and is, importantly, dependent on particle beam densities. Since particle beams are streams of charged particles, high-density beams have a high intensity with a small beam size. Beams are typically classified as either unbunched (or coasting beams), which have no particular shape in the direction of beam motion (longitudinal axis), or bunched, which is when the charged particle stream is distributed to short pulses, each of which are referred to as bunches, containing $\approx 10^{10} - 10^{12}$ particles. Colliders use the latter substructure for beams of particles, assuming a Gaussian distribution within each pulse.

However, unlike p^+p^+ colliders, where there is a direct source that generates proton beams that are then accelerated to collide, muon colliders require a multi-step μ^- production. The diagram of the proposed 10 TeV c.m.e. Muon Collider can be found in Figure 1. A 4 GeV p^+ source generates bunches (pulses) of protons that hit a fixed target material, all within the Proton Complex/Driver, generating hadronic showers from which charged pions ($\pi^\pm$) are produced. These pions subsequently decay into muons, which are then captured and formed into bunches. For true colliders like the $\mu^-\mu^+$ collider (as opposed to fixed-target colliders such as the Spallation Neutron Source, or the fixed-target Proton Complex before the muon accelerator ring in Figure 1), if one has two bunches N_1 and N_2 colliding head-on with one another with frequency f ,

$$\mathcal{L} = f \frac{N_1 N_2}{4\pi\sigma_x^* \sigma_y^*} \quad (3)$$

where σ_x^* is the transverse root-mean-squared width of Gaussian beam sizes [6]. , which will be the focus of the next section, or the proton driver fixed-target system inside the $\mu^-\mu^+$ collider As Equation 1.1 indicates, to obtain a high luminosity, other than increasing the number of particles per bunch, the number of bunches, and the frequency of collisions, one can consider decreasing the rms width of the transverse beam at the interaction point. For the latter to be true, one could either produce particle bunches that are narrowly distributed in spacetime from the start, or focus existing particle bunches and narrow down the transverse beam size that way. In the case of the Muon Collider, high luminosity requires tightly packed $\mu^\pm$ (“compressed”) bunches colliding at the interaction point. However, muons in the Muon Collider originate from the decay of pions, which are, in turn, produced by high-intensity protons colliding with a target. This means that to ensure the resulting pions, and therefore muons, are produced in a spatially compact and temporally coherent way suitable for capture and acceleration, first the proton bunches must be compressed longitudinally before striking the target, causing the pion production to occur within a short time window. In the pion production stage, right at the end of the proton driver of the Muon Collider, the initial timing and the position of the proton impact determine the start of the proton interaction and the subsequent hadronization, creating the pion, allowing for better control of the resulting secondary beam of pions. Then, to further achieve sufficiently dense muon bunches, muons undergo several stages of cooling (ionization cooling, bunching, phase rotation) before being accelerated and collided.

Going back to Figure 1, after the μ Injector Complex shown in (a), the muon bunch goes through a system of linear accelerators (linacs), and two recirculating linacs accelerate the

beams to 60 GeV. This is followed by one or more rings made up of either fast-pulsed synchrotrons or Fixed-Field Alternating gradients (FFA) accelerators. Lastly, the beams are injected at full energy into the collider ring and are stored until they decay.

Given the increased complexity of the setup of the collider as well as the physical considerations, such as higher beam-induced background (unwanted radiation and secondary particles produced indirectly by the beam) or space-charge (discussed in the next section), muon colliders come with unique challenges. The challenge concerning this project begins with the first piece of the muon collider complex, the proton driver, as shown in (b), which consists of a high-power acceleration section, an accumulator, a compressor, and a target delivery section. Generation of a muon beam through the decay of pions produced from proton-target collision requires short, intense proton bunches with a desired bunch length of 2 ns rms width (≈ 0.544 meters). One of the critical steps then is the compression of a relatively long (in about hundreds of ns, 350 ns for Spallation Neutron Source, ≈ 95 meters) initial proton bunch to its rather short final length. This leads to an unprecedented, largely unexplored space-charge regime. The proton driver design and simulation must be able to model the beam dynamics in this regime confidently.

1.2 Longitudinal Beam Dynamics and Proton Bunch Compression: An Introduction

Much of our understanding of beam evolution as the beam goes through the collider relies on accurate analyses of the beam shape and emittance in 4D phase space. We typically restrict our visualizations of the phase space to 2D slices where the horizontal axis is one of the three spatial coordinates. For understanding the beam's longitudinal dynamics, it is typically taken to be the $+z$ direction; for understanding the dynamics in the transverse plane to the beam, this is taken as one of the $+x, +y$ directions. The vertical axis for the transverse directions is the momentum spread $+\Delta p_x, +\Delta p_y$ respectively, while in the longitudinal direction it's the energy spread ΔE . The beam emittance then is the area occupied by the beam in the phase space. An example of bunch distribution in phase space can be seen in Figure 2. However, unlike the beam distribution in the Figure, the beam distribution in both the proposed Muon Collider as well as the Spallation Neutron Source at ORNL is affected by space-charge effects as well as beam instabilities and wakefields. Space-charge is the collective effect of self-induced electric fields caused by non-uniformities in the beams line charge density in high-density beams, as they have densely packed particles

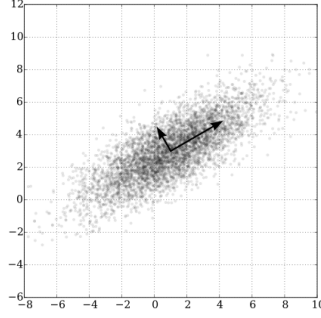


Figure 2: Example beam distribution in phase space [7].

that repel one another. For a round, continuous, uniform-density beam of radius R , we can describe the transverse and longitudinal electric field due to the beam itself inside the beam as:

$$E_{transverse}(r) = \frac{\lambda(z)}{2\pi\epsilon_0} \frac{r}{R^2} \quad E_{longitudinal}(z) = -\frac{Q}{2\pi\epsilon_0\gamma^2} \Lambda \frac{d\lambda(z)}{dz} \quad (4)$$

[8] where $\Lambda = \ln(b/a) + 1/2$ as a is the radius of the beam and b is the beam pipe radius. Here, although both fields lead to a force $F \propto \frac{1}{\gamma^2}$, for short, intense bunches, as we go to ultra-relativistic speeds for the particles, the transverse longitudinal space charge becomes negligible whereas the longitudinal space charge effects distort “orbits” created in phase space by bunches and increases emittance, making bunch compression much more difficult [9]. Additional effects include wakefields, electromagnetic fields generated by a charged particle beam as it passes through accelerator structures (like cavities, beam pipes, or discontinuities), as well as transverse instabilities (beam oscillation in the transverse plane) and longitudinal instabilities (uncontrollable bunch length and energy spread changes).

The spatial compression of a particle bunch length is done through the use of radiofrequency cavities (RF). One starts with an approximately elliptical orbit in the longitudinal phase space where the major axis is horizontal and the minor axis is vertical, i.e., there’s minimal energy spread while there’s a large Δz spread. Radiofrequency cavities apply an oscillating electric field such that the RF voltage can be written as:

$$V(t) = V_0 \sin(\omega_{RF}t + \phi_{sync}) \quad (5)$$

where ω_{RF} is the RF frequency and ϕ_{sync} is the synchronous phase [9] (usually the phase that gives the intended energy gain). The applied electric field acts on the particles in the bunch

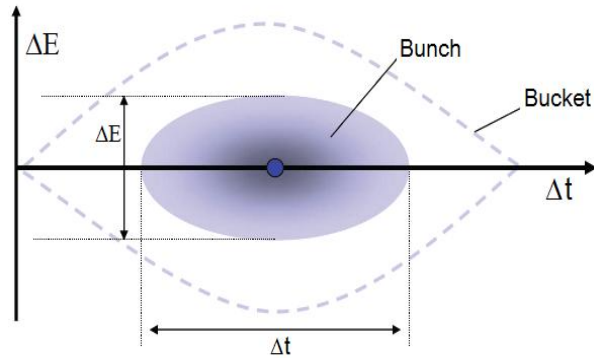


Figure 3: Initial position of the bunch with the RF bucket in longitudinal phase space [10]

such that the particles that have speeds differing from the synchronous particle (particle which arrives at each RF cavity exactly at the designed synchronous phase) have their speeds adjusted about the synchronous particle, i.e. particles that are faster get decelerated and particles that are slow get accelerated. The orbits realized through this action can be seen in Figure 3, where the outermost enveloping orbit is called an RF bucket. This applied electric field thus induces a correlated energy change along the bunch (energy chirp). As particles have different energies, they drift at different speeds, which transforms the energy chirp into a change in bunch length by a 90° rotation from the initial position of the bunch in longitudinal phase space, causing minimal bunch length spread while increasing the energy spread of the particles. Ideally, this new shape retains the same elliptical shape; however, due to longitudinal space-charge effects observed in actual experiments, the shape will likely be smeared across phase space. Finally, the harmonic number h of an RF cavity is the ratio

$$h = \frac{f_{RF}}{f_{rev}} \quad (6)$$

where f_{RF} frequency of the RF accelerating field and f_{rev} revolution frequency of the beam, i.e., how often particles complete one full turn. This determines how many bunches can be stored around the ring, as it defines the number of RF buckets available in the accelerator ring.

2 Spallation Neutron Source and the Experiments

The Spallation Neutron Source at ORNL is uniquely positioned to explore different experiments for a successful proton bunch compression for the desired value of 2 ns because, firstly,

unlike any other currently operational accelerator, SNS successfully achieves the nominal power required from the proton source of a 10 TeV Muon Collider with ≈ 2 MW [3], [11]. It is thought that the SNS accumulator could allow for experimental tests of some beam formation steps of the proton driver, since the beam intensities thus available to SNS, combined with enhancement of the collective effects at its lower energy, may provide space-charge regimes relevant to a muon collider. Novel high-intensity beam technologies such as charge-exchange injection and space-charge mitigation techniques [11] can also be demonstrated at SNS, which potentially already has components of the required proton driver and muon ionization cooling demonstrators to build for a muon collider. Finally, SNS is well-equipped in terms of diagnostics to fully characterize the resulting bunch distribution and already has the simulation tools to benchmark beam tests and experimental schemes for the desired rms bunch width through the pyORBIT package, a Python wrapper of ORNLs ORBIT code that aims to model SNS [12]. With this in mind, this section will outline SNS as well as the project.

2.1 Overview of SNS

An overview of the Spallation Neutron Source can be found in Figure 4. The SNS has four RF cavities; RF1 and RF2 are $h = 1$ cavities whereas RF3 and RF4 are $h = 2$ cavities. The data from the experiments are obtained in the form of current from the Beam Current Monitor in units of Amperes.

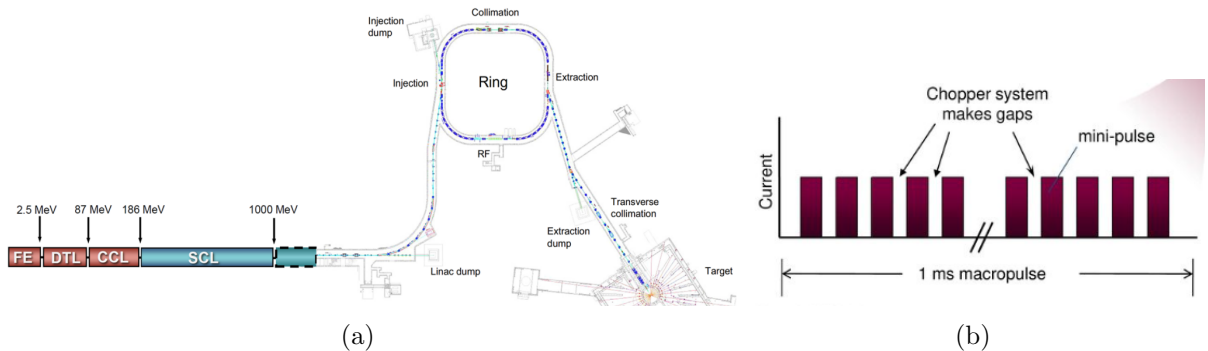


Figure 4: (a) SNS overview where the proton beam coming from the source is “chopped” into $1\mu\text{s}$ pulses (i.e., bunches) within a 1 ms train, also called 1 turn, (shown in (b)) before entering the linac, i.e. being “injected” into the ring, on the left of the Figure. The linac comprises FE: Front-End Injector, DTL: Drift Tube Linac, CCL: Coupled Cavity Linac, SCL: Superconducting Linac, accelerating bunches in steps before they are accumulated in a ring and compressed in the same ring via the RF at the bottom of the ring [13].

2.2 Project and Goals

The main goal of this project is to study the collective effects in the extreme space charge regime found in high-intensity accelerators like MuC by looking at compression of 350 ns rms width p^+ bunches in SNS. On the experimental side, initial data on various beam accumulation, bunch compression, and beam storage experiments in the SNS Accumulator Ring at ORNL were completed. On the simulation side, pyORBIT was used to simulate and cross-compare with experimental data for benchmarking as well as adding in the observed physical effects in experiment at the University of Chicago. For unmeasurable free parameters, the goal was to optimize them via gradient descent algorithms to fit observed shapes in the experiment.

The first experiment, conducted in January, was simply to gather data when the beam in the ring is accumulated with RF cavities turned off and then stored in the ring when RF $h = 1$ cavities were turned on. Due to the ferrite material of the RF cavities and the danger of overloading, 10 kV of amplitude was selected as the maximum voltage supplied to the cavities. For maximum bunch compression, a bunch accumulation with uniform density filling of $\approx 2/3$ of the ring (in the experiment, labelled as PWON, typically actually about $\approx 36/64$) for the initial spatial spread was desired with ideally minimal energy spread, so that after the RF rotation, the Δz spread would be minimal. It was expected that, without the RF cavities, the absence of external longitudinal forces would allow for the particles to fill almost the entire ring, but, in return, it would keep the initial energy spread minimal. It was then the goal to see if it would be possible for the RF cavities to rotate the bunches efficiently before the bunches filled the entire ring and decohered. Unfortunately, this experiment was conducted right after a long maintenance period for the ring, which meant some unexpected yet unphysical results were observed. Because of this, the data from this experiment was not used in the final results.

The second experiment, or series of experiments, was conducted on February. The purpose of the experiments listed in Table 1 was to make sure the relatively simple experimental schemes would correctly be benchmarked in subsequent pyORBIT simulations. In Experiment 02, for all cases, the beam is injected for a short time (i.e. 100-200 turns), and then it is stored for hundreds of turns, all with a constant $h = 1$ RF drive voltage. It is expected that if the RF voltage is weak, the phase space distribution will not rotate a significant amount during injection; after which, a moderate bunch compression is expected at ≈ 500 turns. While this paper will benchmark only Experiments 01 and 02, other Experiments that have

been conducted throughout the beam time are also listed. For Experiment 03, the second cavity was turned off immediately after accumulation to mimic the instantaneous turn-on of a single 5 kV cavity, and there was a 180° phase difference between the 2 RF cavities. For Experiment 04, the RF1 Phase was 118° , the RF2 Phase was 100° , and they were operated with some delay in ramping up the voltage (about 200 turns).

Experiment	Case	State of RF1 RF2 RF3 RF4	Injection Turns	Total Turns
Experiment 01	N/A	All RF Cavities OFF	300	1000
Experiment 02	Case 01	RF1 = 10 kV	100	1200
Experiment 02	Case 02	RF1 = 10 kV	200	1200
Experiment 02	Case 03	RF1 = 10 kV	300	1200
Experiment 03	Case 01	RF1 = 5 kV RF2 = 5 kV	100	900
Experiment 04	Case 01	RF1 = 10 kV RF2 = 10 kV	50	900

Table 1: Experiments conducted at SNS Control Room.

Parameter	Value
p^+ Energy	1.3 GeV
ΔE	0.001 GeV
h=1 RF Voltage	RF1 = 10 kV, RF2 = 0 V
Macroparticles per turn	3000

Table 2: Simulation parameters for pyORBIT

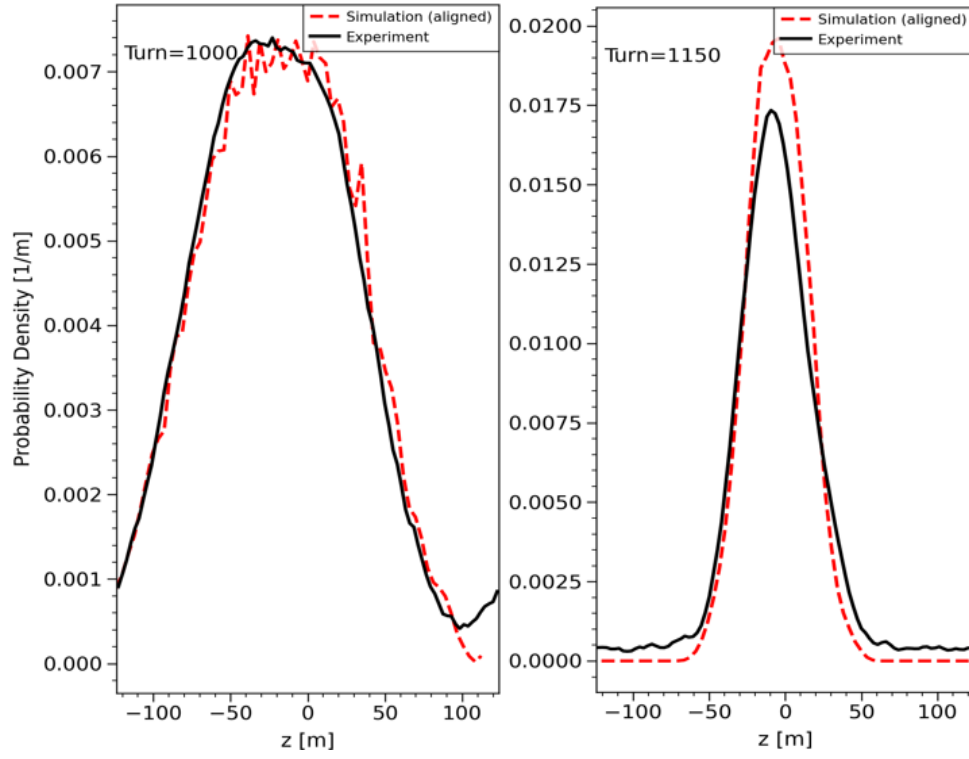
On the simulation side, the Table of parameters 2 was used to simulate Experiments 01 and 02 through pyORBIT. pyORBIT utilizes 3D Particle-in-Cell (PIC) methods and Fast Fourier Transform (FFT) Poisson solvers to accurately model space charge forces. It simulates both longitudinal and transverse beam dynamics with Gaussian and JOHO distributions [14] respectively. The simulation first inserts 3000 macroparticles, i.e. about $3 \times 10^9 - 3 \times 10^{12}$ particles each turn instead of simulating every particle in a bunch. Then the energy of the synchronous particle is initialized as well as an initial energy spread for the bunch and the synchronous particle is tracked turn by turn through the SNS Ring lattice, which includes key elements such as beam position monitors, DRIFT (drift region, where there is no external electromagnetic fields), COL (the collimator, a device with apertures or absorbers that defines beam size or shape and removes halo or stray particles), OCT (octupole magnets, which apply a nonlinear magnetic field to correct higher-order aberrations or manipulate the

beams phase space), Q (quadrupole magnets, which focus the beam in one transverse plane and defocus it in the other.), DH/DB (dipole horizontal/dipole bending magnets, which are used to bend or steer the beam trajectory, especially in the horizontal plane), IKICK (kicker magnet, which is a fast pulsed magnet that deflects the beam into or out of a ring) [13]. This is also where the RF cavities apply an electric field to the simulated particles. Then, the phase space diagram discussed in the previous section, as well as probability density vs. spatial spread graphs for every turn, and the rms width of the bunch vs. turn graphs, will be plotted, as discussed further in the next section. In addition to the effects discussed above, longitudinal space charge effects were included through adding a longitudinal space-charge node developed for the SNS [15].

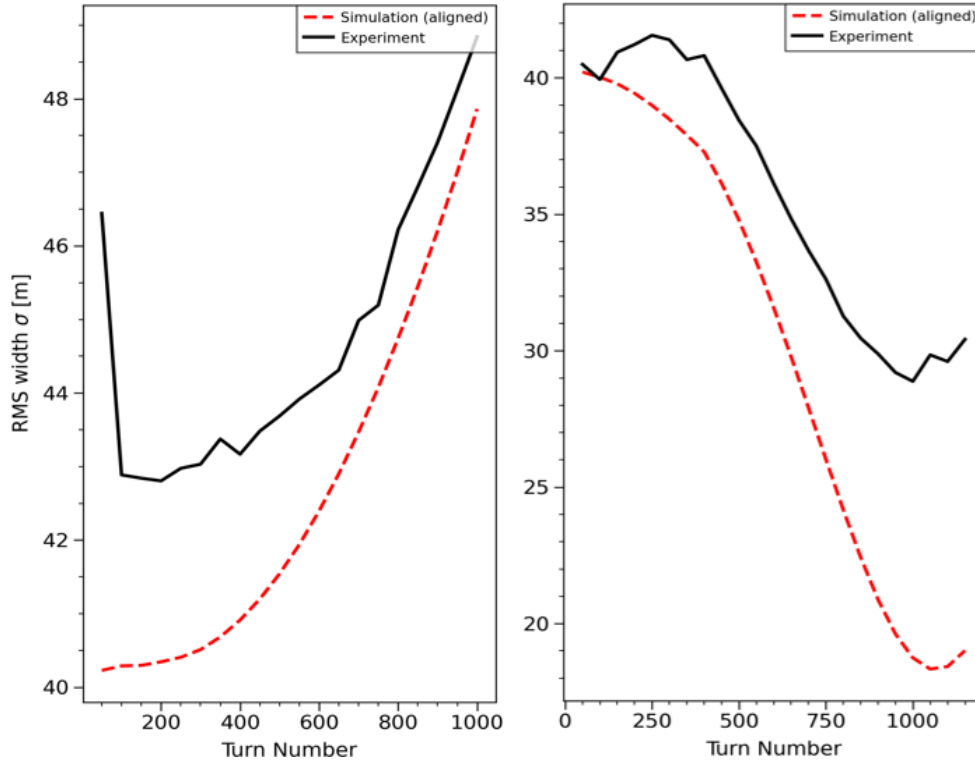
BFGS gradient descent algorithm [16] was used to match the simulation rms width to the experimental rms width to determine the energy of the synchronous p^+ as well as the energy spread of the particles ΔE in the experiment as they are typically parameters that cannot be directly measured in the experiment.

3 Results and Discussion

The main goals of this project were thus 1) to determine the parameters and physical effects involved in the proton bunch compression experiments conducted at SNS and 2) to successfully decrease the proton bunches from its starting point 350 ns within the 1000-1200 turns that each experiment ran for. To this end, the probability distribution versus the spatial spread of the bunches in turn 1000 for Experiment 01 and turn 1150 for Experiment 02 as well as the rms widths (in meters) versus the number of turns for Experiment 01 and Experiment 02 have been plotted to see whether if these goals have been achieved in Figure 5. Converting the rms widths to seconds, for Experiment 02, the experiment for which RF cavities were actively used for bunch compression, the experimental range is $\approx 106 - 154$ ns whereas the range predicted by the simulation is $\approx 66 - 146$ ns, both of which are significant improvements from the starting point. One can then observe that while the probability density of Experiment 01 is accurately mirrored by its corresponding simulation data, the numerical instabilities at the peak of the probability density graph are still causes of concern. This is supported by the rms width graph for Experiment 01, as while the overall shape is matched by the simulation (note the 0 turn point in the experiment is likely due to low statistics as there are very few to no particles injected during turn 0) there is still discrep-

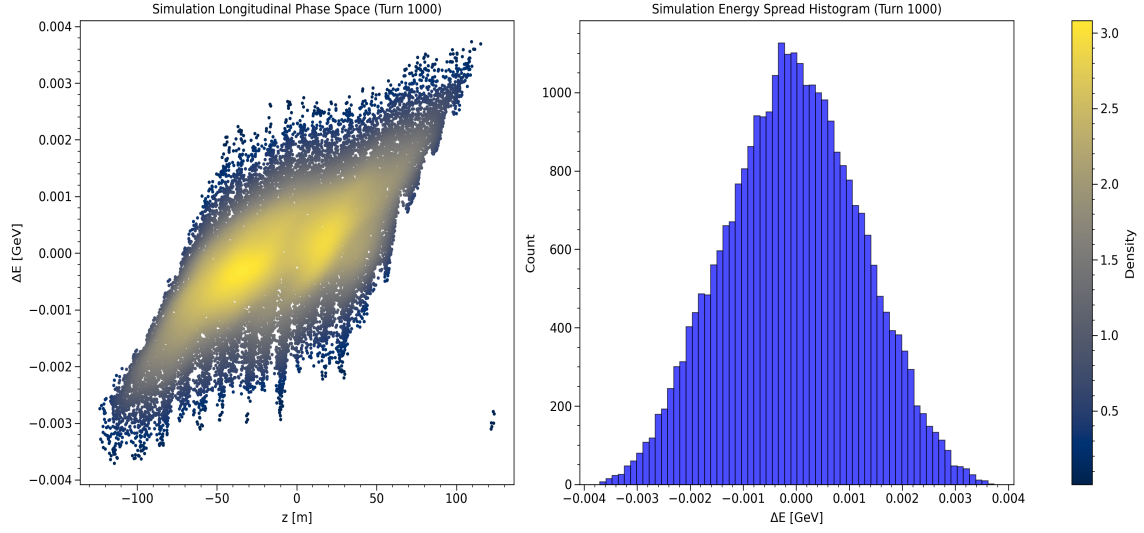


(a)

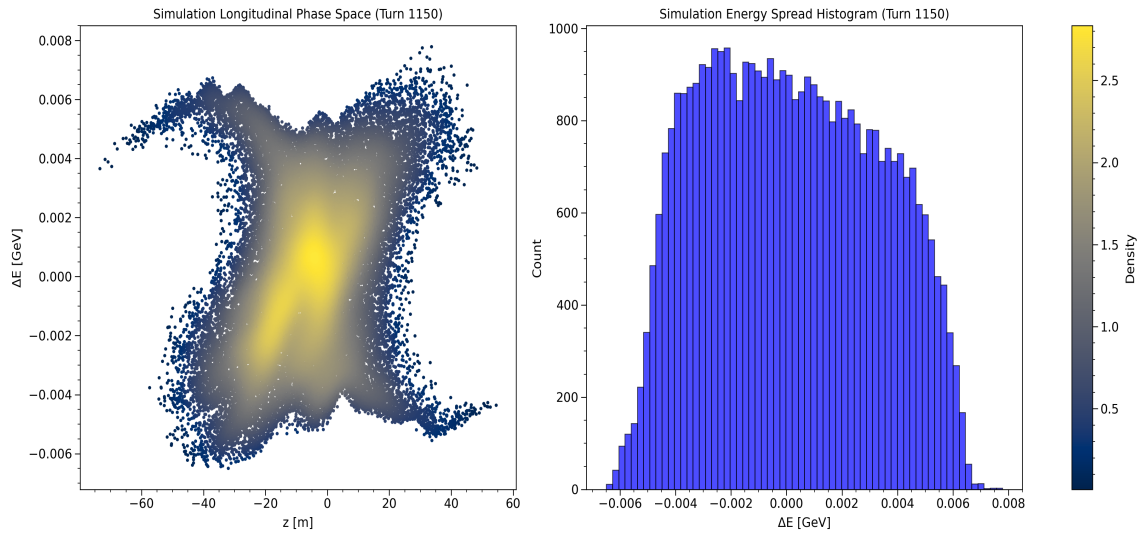


(b)

Figure 5: Simulation and Experimental Results for Experiment 01 (left) and Experiment 02 Case 03 (right) of longitudinal spread after turn 1000 (a) and RMS width in meters v. Turns (b).



(a)



(b)

Figure 6: Simulation Results for Experiment 01 (a) and Experiment 02 Case 03 (b) of the phase space of the particles after turn 1000 (a) turn 1150 (b).

ancy between the rms widths between the simulation and experiment. The gap between simulation and experiment is larger for Experiment 02 since while the probability density shows a narrow peak for both experiment and simulation, thus indicating a narrowing in the spatial spread (or “bunching”), the experimental data seem to indicate greater dilution in the probability amplitude at $z = 0$ than the simulation. In the case of Experiment 02, the shape is similarly matched, however, the simulation seems to predict a much greater compression of the bunch length by turn 1150, which could be an indicator that we need to consider additional physical effects not currently present in pyORBIT that prevent bunch length compression and facilitate bunch shape distortion.

Figure 6 shows phase diagrams of turns 1000 and 1150 for Experiments 01 and 02 (Case 03) respectively for 100 macroparticles per turn in order to display the probability densities of the locations in the phase space as the third (color) axis. From this we can see that while the drifting of particles and the distortion of the elliptical shape in Experiment 01 make sense as we are not controlling particle speed through RF cavities, the phase space in Experiment 02 is very clearly distorted due to longitudinal space charge effects as the spatial spread decreases much less than the decrease one would see just through 90° rotation done by RF cavities. The numerical instabilities in simulation observed in Figure 5 for Experiment 01 similarly show up in Figure 6(a) as the edges have a zigzag pattern. This is not observed in simulations for Experiment 02 and the distortion from longitudinal space charge is apparent.

As a result, one could conclude that at the end of this project:

1. Bunch compression stronger than that of baseline operations in the SNS ring was demonstrated, indicating the likelihood of being able to further compress the bunch for MuC studies using the ring, even before IMCC proposed hardware updates such as adding another ring.
2. It was shown there are limitations in the physical effects accounted for in pyORBIT that need to be addressed. This includes numerical instabilities in 1D space-charge (3D needed) as well as a model of the RF system that takes beam loading into account, higher complexity beam-cavity interactions.
3. Further experiments changing the timing, relative phase difference, feedback, and voltage of RF cavities as well as number of injection turns were conducted. These could give new insight into the seemingly missing physical effect in the pyORBIT simulation for Experiment 02 Case 03.

Acknowledgements

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